

A
PROJECT REPORT
ON
Library Management System
PROJECT REPORT SUBMITTED IN THE
PARTIAL FULFILMENT FOR THE
DEGREE OF
BACHELOR OF SCIENCE IN
INFORMATION TECHNOLOGY

SUBMITTED
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BATCH (2021-24)

UNDER THE GUIDANCE OF:
DR.SUNIL KUMAR



AURO UNIVERSITY
SURAT

Academic Year 2023-24

CERTIFICATE OF UNDERTAKING

I, **Yagnik Narola** Enrollment No.**212021029** hereby declare that project entitled Logistic Management undertaken at School of Information Technology for the partial fulfilment of BSc. IT degree. This Project is my original work and the Project has not formed the basis for the award of any other degree, associate ship, fellowship or any other similar titles, either in AURO University or any other University of India.

YAGNIK NAROLA

SCHOOL OF INFORMATION TECHNOLOGY CERTIFICATE

This is to certify that the project entitled **BookWise** has been carried out by the student under my guidance in the partial fulfilment of the Degree of **Bachelor of Science in Information Technology** of AURO University, Surat during the academic year 2021-24 (Semester-VI).

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Internship Letter

Mr Yagnik Narola,

We are delighted to inform you that you have been selected for the internship position at Bitrix Infotech PVT LTD, for the role of intern Node JS developer. We are confident that your skills and passion will make a valuable contribution to our company.

Your skills and qualification stood out among a competitive pool of candidates, and we believe you will contribute significantly to our team. Your internship duration will start from 02-01-2024 to **30-03-2024** and your internship hours will be 9:00 AM to 6:00 PM. **Sr Node JS Developer Viny Shah** will assist throughout the internship.

We look forward to welcoming you to our team and are confident that your time with us will be both rewarding and enriching.

Congratulations, and welcome to the Team Bitrix Infotech PVT LTD



Your Sincerely,

Aayush Soneji | HR Manager
Bitrix Infotech PVT LTD.



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Lal Darwaja Station Road, Surat, Gujarat, India, 395003

Acknowledgement

First and foremost, we extend our gratitude to the team at BITRIX INFOTECH, particularly Mr. VINY SHAH, for entrusting us with the opportunity to bring our vision to life under their guidance. Their expertise and encouragement have been invaluable throughout the development process, fostering an environment conducive to learning and innovation.

We are indebted to Dr. Sunil Kumar, our mentor and guide, whose unwavering support has been the cornerstone of our project. Dr. Kumar's insights and feedback have not only shaped the direction of our app but have also instilled in us a sense of diligence and determination to overcome challenges.

Our heartfelt thanks also go to the faculty and staff of the Department of Information Technology for their continual support and encouragement. Their collective efforts have provided us with a nurturing environment in which to explore and expand our capabilities.

To our friends and family who have stood by us during the highs and lows of this journey, we express our deepest appreciation. Your encouragement and understanding have been a source of strength and motivation.

Lastly, to all those whose names may not be mentioned here but have contributed in their own way, we extend our sincere thanks. Your support, whether big or small, has played a vital role in the realisation of our app project.

Thank you, one and all, for being a part of this remarkable journey.

Thanking All,
Yagnik Narola

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INTRODUCTION

1.1 Project Overview:

Our final project is the culmination of efforts to develop a contemporary Library Management System featuring a dual-panel interface tailored for students and library staff. This comprehensive system optimises book requests, refines resource management, and ensures a smooth user experience.

1.2 Purpose:

The primary purpose of this project is to simplify the process of borrowing books for students and improve the book management process for library staff. We aim to leverage the capabilities of Flutter for the frontend development, creating a user-friendly and responsive interface. Firebase, a powerful cloud-based platform, will be employed as the backend database to ensure real-time data storage and synchronisation.

1.3 Scope:

Student Panel:

- Allows students to register, log in, and request books from the library's collection.

Library Staff Panel:

- Empowers library staff to manage the library's inventory by adding, editing, or removing books. They can also oversee and respond to student book requests.

Firebase Database:

- Ensures that all data, including user accounts, book records, and request statuses, is securely stored, and updates are instantly reflected across all panels.

1.4 System Panel:

Student Panel

Description:

- The Student Panel serves as the user interface for students who are registered users of the library management system. Its primary function is to provide students with a user-friendly platform for accessing library services, searching for books, and requesting books.

Key Features:

- User Registration and Login : Students can create an account by providing their details and then log in securely to access the system.
- Book Search : Students can search for books by title. This feature helps them find the books they need efficiently.
- Book Request : Students can request books they wish to borrow. When a book is requested, the system records the request.
- Logout: Students can securely log out of their accounts when they are done using the system.

Librarian Panel

Description:

The Librarian panel is equipped with features to streamline book management and ensure efficient communication with students.

Key Features:

- **User Authentication:** Librarians can securely log in to access the panel with administrative privileges.
- **Book Management:** Librarians can add new books to the library's collection, edit existing book records .
- **View Student Requests:** Librarians can see a list of student book requests, including details about the requested books and the students making the requests.
- **Approve/Decline Requests:** Librarians have the authority to approve or decline student book requests based on the availability of books or library policies.
- **Manage Borrowing:** Librarians can keep track of books that are checked out by students, due dates, and issue reminders or penalties for overdue books.
- **User Management:** This feature allows librarians to manage student accounts, including account creation, updating user information, and account deactivation if needed.
- **Logout:** Librarians can securely log out of their accounts when their administrative tasks are completed.

These two panels work in tandem to create an efficient library management system, with students using their panel to request books, and librarians using their panel to manage books and student requests. The system ensures a smooth and organised process for borrowing and returning library resources

1.5 OBJECTIVES

- **Requirement Gathering and Analysis:**

- Conduct interviews and surveys with stakeholders to gather detailed requirements for both student and librarian panels.
- Create user stories and personas to understand user needs and preferences.
- Analyze gathered information to define project scope, objectives, and key features required for each panel.

- **System Design:**

- Develop a comprehensive design document outlining the architecture, components, and interactions of the library management system.
- Create wireframes and UI mockups using design tools like Figma or Adobe XD to visualise the user interfaces for both student and librarian panels.
- Ensure the design aligns with modern UI/UX principles and provides an intuitive user experience for all functionalities.

- **Database Design and Implementation:**

- Define a normalised database schema that efficiently represents entities such as user accounts, book records, requests, and transactions.
- Implement the database using Firebase Firestore, ensuring proper indexing and security rules for real-time data storage and access.
- Incorporate data modelling techniques to optimise database performance and scalability while adhering to Firebase's limitations and best practices.

- **Integration of Front-end and Back-end:**

- Develop the front-end of the system using Flutter, leveraging its widget-based architecture and hot reload feature for rapid development.

- Integrate Firebase Authentication for user authentication and authorization, ensuring secure access to the system for both students and librarians.

- Implement data synchronisation between the front-end Flutter application and the Firebase Firestore backend to enable real-time updates and notifications.

- **Custom Functions & Validations:**

- Implement custom functions for user registration, login, password reset, and account management using Firebase Authentication SDK.

- Develop validation logic for user inputs, book requests, and other forms to ensure data integrity and prevent erroneous submissions.

- Incorporate error handling mechanisms to provide informative feedback to users in case of validation failures or server errors.

- **Making Responsive Design:**

- Design responsive user interfaces that adapt to various screen sizes and orientations, including smartphones, tablets, and desktop browsers.

- Utilise Flutter's layout widgets like MediaQuery, Expanded, and Flexible to create fluid UI designs that scale gracefully across different devices.

- Conduct thorough testing on different devices and screen resolutions to ensure consistent and optimal user experience across all platforms.

- **Testing:**

- Perform functionality testing to verify that all features and interactions work as intended according to the requirements.

- Conduct usability testing with representative users to gather feedback on the user interface design and navigation flow.

- Perform security testing to identify vulnerabilities such as injection attacks, data leaks, and unauthorised access attempts, and apply necessary safeguards.

- **Debugging and Code Optimization:**

- Debug and troubleshoot issues reported during testing, using debugging tools and logging mechanisms to identify root causes and implement fixes.

- Optimise codebase for performance by reducing redundant computations, minimising network requests, and implementing caching strategies where applicable.

- Conduct code reviews with team members to ensure adherence to coding standards, best practices, and maintainability guidelines.

- **Documentation:**

- Create comprehensive project documentation covering requirements, design decisions, implementation details, and testing procedures using Google Docs or similar tools.

- Prepare user manuals and guides for both students and librarians, providing step-by-step instructions on how to use the system effectively.

- Ensure documentation is organised, up-to-date, and easily accessible to stakeholders for reference and future maintenance.

- **Final Review and Submission:**

- Conduct a final review of the entire project to ensure all objectives have been met, and the system is ready for deployment.

- Address any outstanding issues or feedback identified during the review process, making necessary adjustments to achieve project completeness and quality.

- Submit the finalised project report and deliverables to the internal guide for evaluation, ensuring all requirements and expectations are met before final submission.

Literature Review

2.1 Review of Existing Library Management Systems:

Existing library management systems serve as the foundation for understanding the landscape and identifying opportunities for improvement. Various systems, both commercial and open-source, offer a range of features such as cataloguing, circulation, and patron management. Examples include Koha, Evergreen, and Alma, each with its strengths and weaknesses. Koha, an open-source integrated library system (ILS), is renowned for its robust cataloguing capabilities and community-driven development model. Evergreen, another open-source ILS, emphasises scalability and flexibility, making it suitable for consortial environments. Alma, a cloud-based system offered by Ex Libris, stands out for its comprehensive suite of features and seamless integration with other library services.

However, these systems also face common challenges such as complex user interfaces, limited customization options, and high implementation costs. Moreover, emerging trends like mobile access, data analytics, and AI-driven recommendations present new opportunities for innovation in library management systems. By studying these existing systems and understanding their strengths and weaknesses, we can identify areas for improvement and innovation in our project.

2.2 Technologies and Frameworks Used:

The choice of technologies and frameworks significantly impacts the development and functionality of the Library Management System. In our project, we have selected Flutter as the primary development framework due to its cross-platform capabilities, fast development cycle, and rich ecosystem of plugins. Flutter allows us to build native-like mobile applications for both Android and iOS platforms, ensuring a consistent user experience across devices.

For the backend infrastructure, we are leveraging Firebase, a comprehensive platform provided by Google, offering features such as authentication, real-time database, and cloud storage. Firebase

enables us to build scalable and reliable backend services with minimal setup and configuration, allowing us to focus on building core features of the system.

Additionally, we are exploring other technologies such as GraphQL for efficient data fetching, and Material Design for creating visually appealing and intuitive user interfaces. By carefully selecting and integrating these technologies, we aim to create a modern and efficient Library Management System that meets the needs of both students and library staff.

2.3 Analysis of Key Features and Functionalities:

The success of a library management system depends on its ability to fulfil key user requirements and provide essential functionalities. Through an analysis of existing systems and user feedback, we have identified several key features and functionalities that are crucial for our project:

- **User Authentication and Authorization:** Secure authentication mechanisms to ensure only authorised users can access the system and perform relevant actions.
- **Book Search and Discovery:** Advanced search capabilities to enable users to easily find books based on title, author, genre, or keywords.
- **Book Borrowing and Returning:** Seamless processes for students to request loans, receive notifications, and return books within specified deadlines.
- **Inventory Management:** Tools for librarians to manage the library's collection, including adding new books, updating records, and removing outdated items.
- **Notifications and Alerts:** Real-time notifications to keep users informed about due dates, overdue fines, and other important events.

Environment Design

3.1 Hardware and Software Requirements:

Hardware Requirements:

- **Client Devices:** The Library Management System is designed to run on various client devices, including smartphones, tablets, laptops, and desktop computers. Minimum hardware specifications include:
 - **Processor :** Any modern processor capable of running the operating system and supporting Flutter applications.
 - **Memory (RAM) :** At least 2GB of RAM for optimal performance.
 - **Storage :** Sufficient storage space for storing the application and local data.

Software Requirements:

- **Operating System:** The application supports multiple operating systems, including:
 - **Android :** Android 4.1 (Jelly Bean) or later versions.
- **Flutter SDK:** The latest version of the Flutter SDK installed on the development machine for building, testing, and deploying Flutter applications.
- **Integrated Development Environment (IDE):**
- **Recommended IDEs include:**

- Visual Studio Code : With Flutter and Dart plugins/extensions for coding and debugging Flutter applications.
- Android Studio : With Flutter and Dart plugins for native Android development and Flutter support.
- **Firestore SDK:** Integration of the Firestore SDK for Flutter to access Firestore services such as Authentication, Realtime Database, Cloud Firestore, Cloud Storage, and Cloud Messaging.

3.2 Server-side and Client-side:

Server-side:

- **Firestore Backend:** The server-side component of the Library Management System is managed by Firestore, a cloud-based platform provided by Google.
 - Firestore Authentication : Handles user authentication and authorization securely.
 - Firestore Realtime Database or Cloud Firestore : Stores and synchronises data related to user accounts, book records, borrowing history, and request statuses in real-time.
 - Firestore Cloud Storage : Stores multimedia assets such as book covers and user profile pictures.
 - Firestore Cloud Functions : Provides serverless functions to execute backend logic, process data, and trigger actions based on events.

Client-side:

- **Flutter Framework:** The client-side component of the Library Management System is developed using the Flutter framework, an open-source UI toolkit created by Google.
 - Dart Programming Language: Flutter applications are written in Dart, a language optimised for building fast and efficient user interfaces.
 - Material Design: Utilises Google's Material Design principles and components for designing visually appealing and intuitive user interfaces.

Methodology

4.1 Overview of the Development Process:

The development process of our Library Management System is driven by a systematic approach that emphasizes collaboration, transparency, and adaptability. It begins with comprehensive planning and requirements gathering, where project stakeholders define the scope, objectives, and success criteria. This initial phase involves conducting stakeholder interviews, analyzing user needs, and prioritizing features to ensure alignment with project goals.

Following requirements gathering, the project enters the iterative design phase, where wireframes, prototypes, and user interface mockups are created to visualize the user experience and validate design decisions. Design iterations are based on user feedback, usability testing, and iterative refinement to optimize the user interface for intuitiveness and accessibility.

Once the design phase is complete, development activities commence, with cross-functional teams working collaboratively to implement features, functionality, and backend integration. Adopting an Agile development methodology, we break down project requirements into user stories and tasks, which are prioritized and scheduled for implementation within short development cycles known as sprints.

Throughout the development process, continuous testing and quality assurance activities are conducted to identify and address defects, ensure adherence to requirements, and maintain product quality. Automated testing frameworks, unit tests, integration tests, and user acceptance testing are performed to validate system functionality and performance.

Upon completion of development, the system undergoes rigorous testing and validation to ensure that it meets the defined acceptance criteria and user expectations. Feedback from stakeholders and end users is collected and incorporated into subsequent iterations, allowing for continuous improvement and refinement of the product.

4.2 Selection of Flutter as the Development Framework:

After extensive research and evaluation of cross-platform development frameworks, we concluded that Flutter was the optimal choice for building our Library Management System. Several factors contributed to this decision:

- **Versatility:** Flutter's ability to target multiple platforms, including Android, iOS, web, and desktop, allows us to reach a broader audience with a single codebase.
- **Performance:** Flutter's reactive framework and compiled native code result in fast and smooth performance, ensuring a seamless user experience across devices.
- **Developer-friendly features:** Flutter's hot reload feature enables rapid iteration and debugging, reducing development time and accelerating time-to-market. Additionally, Flutter's rich set of UI components and libraries simplifies the implementation of complex user interfaces, enhancing developer productivity and creativity.

4.3 Tools and Resources Used:

Our development process leverages a suite of tools and resources to streamline development, collaboration, and deployment tasks. These include:

- **Version Control Systems:** We utilize Git for code management, allowing for efficient collaboration, version control, and code review processes. GitHub serves as our project hosting and collaboration platform, providing features such as issue tracking, pull requests, and project boards to facilitate team communication and coordination.
- **Continuous Integration/Continuous Deployment (CI/CD) Pipelines:** We implement CI/CD pipelines to automate testing, build, and deployment processes, ensuring consistent and reliable delivery of updates and enhancements to end users. Tools such as Jenkins or GitLab CI are used to configure and orchestrate CI/CD workflows,

enabling seamless integration with version control systems and cloud platforms.

- **Integrated Development Environments (IDEs):** Our development team utilizes IDEs such as Visual Studio Code and Android Studio for coding, debugging, and project management. These IDEs provide features such as syntax highlighting, code completion, and integrated debugging tools, enhancing developer productivity and efficiency.

4.4 Project Management Approach:

To ensure project success, we adopt an Agile project management approach characterized by iterative development, frequent communication, and continuous feedback loops. Our project management approach includes the following key practices:

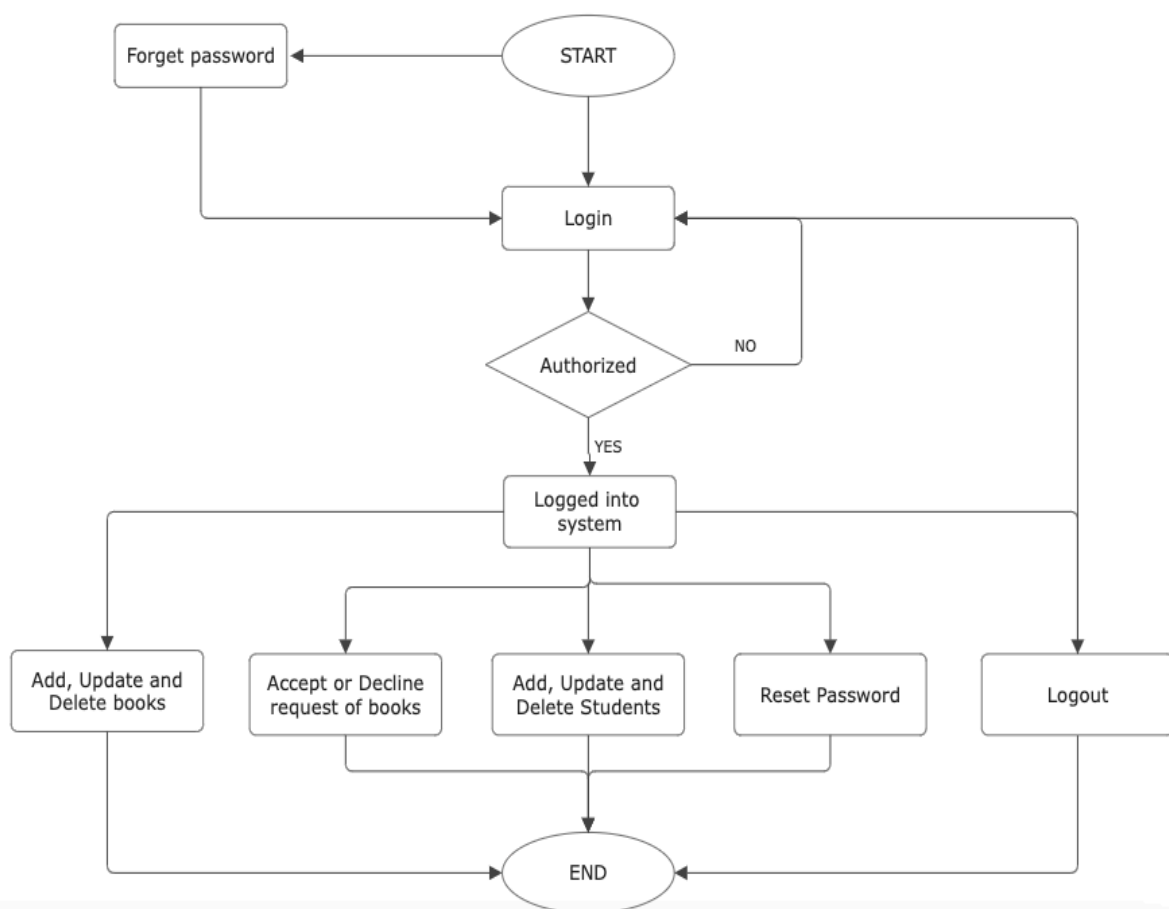
- **Sprint Planning:** We organize work into short development cycles known as sprints, with each sprint delivering a potentially shippable increment of functionality. Sprint planning sessions involve selecting and prioritizing user stories and tasks based on project objectives and stakeholder priorities.
- **Daily Stand-up Meetings:** We conduct daily stand-up meetings to synchronize team activities, discuss progress, identify potential roadblocks, and prioritize tasks for the day. These meetings foster transparency, collaboration, and accountability among team members.
- **Retrospective Sessions:** At the end of each sprint, we hold retrospective sessions to reflect on the sprint's outcomes, identify areas for improvement, and discuss actionable insights and lessons learned. Retrospectives promote continuous improvement and adaptation, enabling us to refine our processes and practices over time.

System Architecture

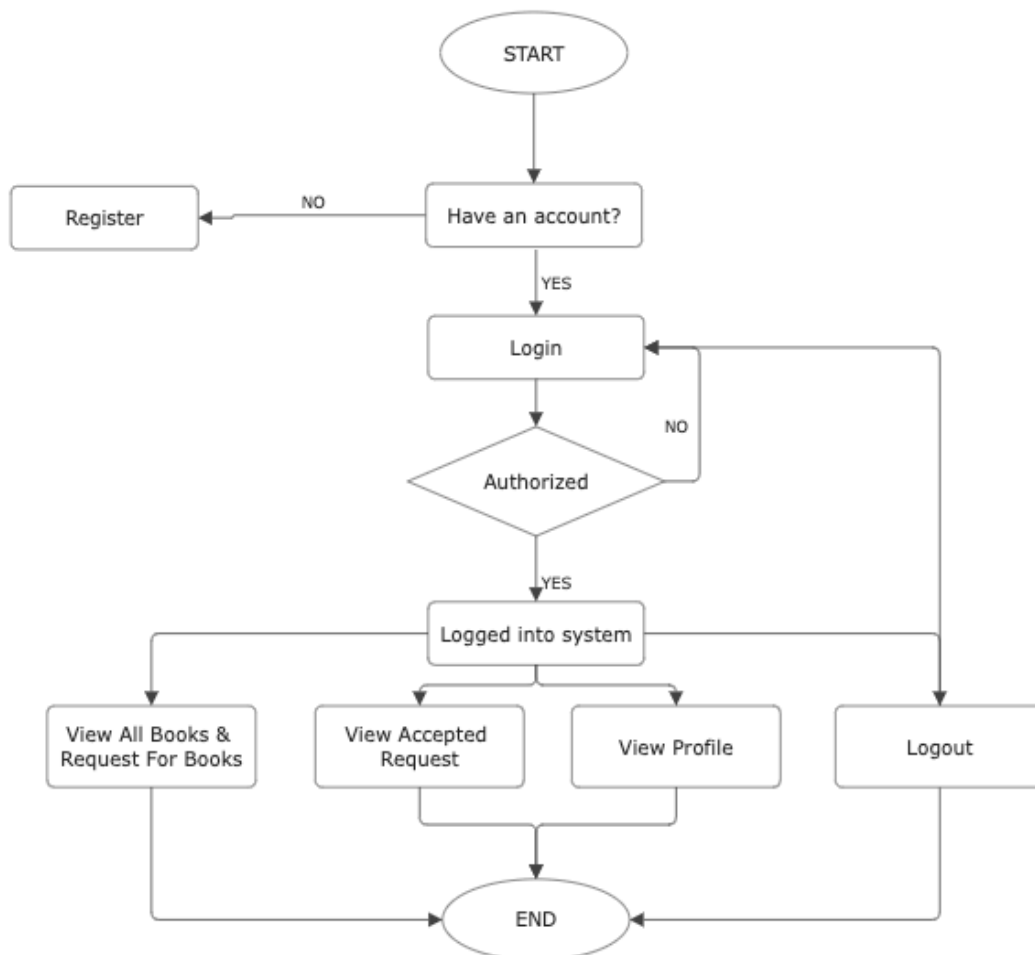
5.1 Architecture Diagram

SYSTEM FLOW DIAGRAM

❖ Librarian Process Flow Diagram:

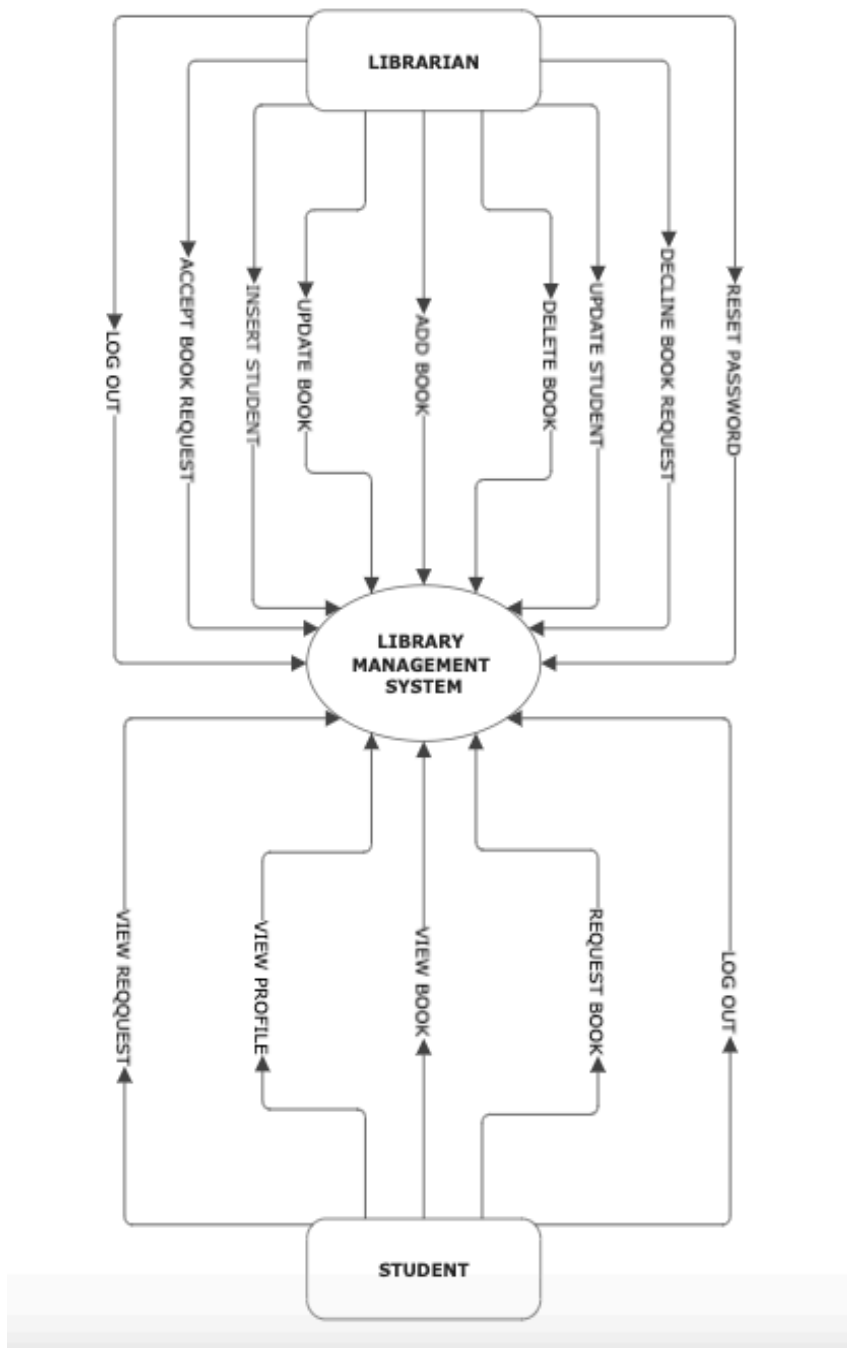


❖ Student Process Flow Diagram:

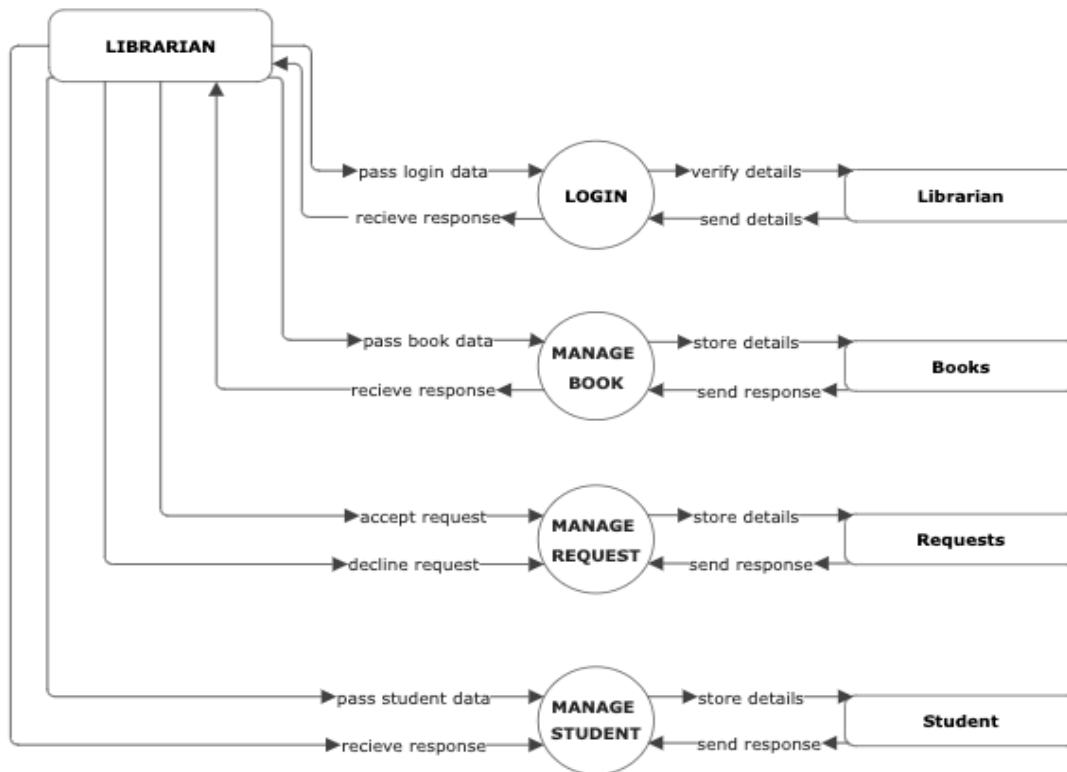


DFD/UML DIAGRAM

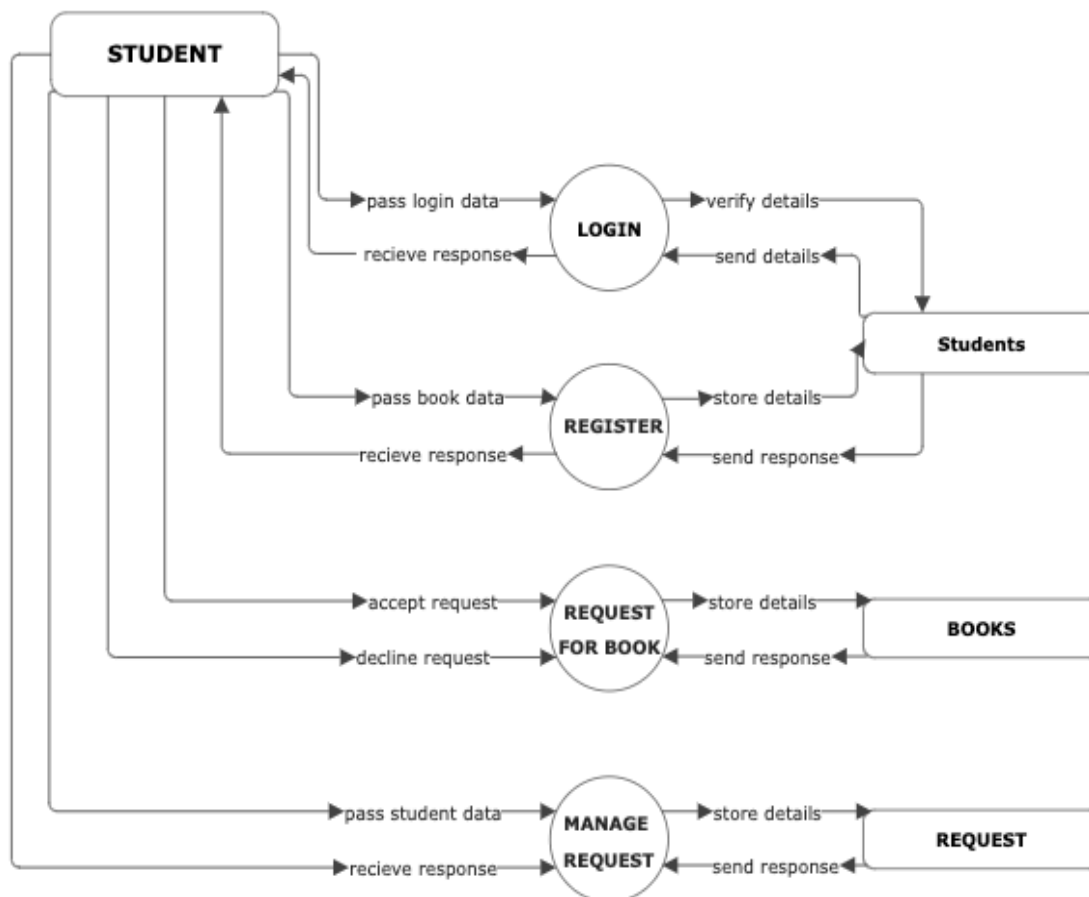
❖ 0 Level Diagram (Context Diagram):



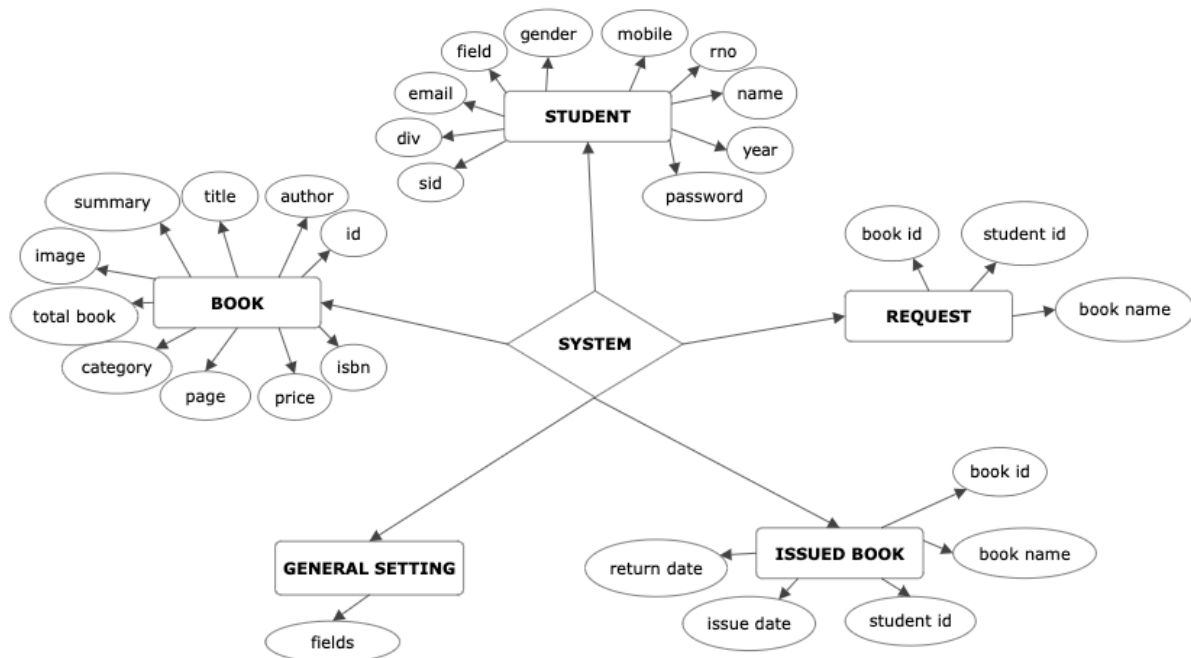
- **1st Level Librarian Diagram:**



- **1st Level Student Diagram:**



ER DIAGRAM



5.2 Components and Their Functionalities

1. User Interface (UI):

- Student Panel: Developed using Flutter widgets to create an intuitive and responsive interface for students to interact with the library system.
- Librarian Panel: Also built with Flutter, providing librarians with administrative tools and functionalities tailored to their
- role.

2. Authentication System:

- Utilises Firebase Authentication to handle user authentication and authorization securely.

- b. Supports authentication methods such as email/password, phone number, Google Sign-In, and others provided by Firebase Authentication.

3. Real-time Database:

- a. Leverages Firebase Realtime Database to store and retrieve data related to user accounts, book records, borrowing history, and request statuses.
- b. Utilises Firebase's real-time synchronisation capabilities to ensure that data changes are instantly reflected across all connected clients.

4. Cloud Storage:

- a. Utilises Firebase Cloud Storage to store and manage multimedia assets such as book covers, author photos, and user profile pictures.
- b. Enables efficient storage and retrieval of large files while ensuring scalability, reliability, and accessibility.

5. Firestore:

- a. Utilises Cloud Firestore, Firebase's NoSQL document database, for storing structured data and performing complex queries.
- b. Stores metadata related to books, users, borrowing history, and other system entities in Firestore collections and documents.

6. Push Notifications:

- a. Integrates Firebase Cloud Messaging (FCM) to send push notifications to users' devices regarding due dates, overdue fines, book availability, and request statuses.
- b. Enhances user engagement and communication by providing timely updates and reminders directly to users' devices.

5.3 DATABASE DESIGN

- **Book Table:**

No	Field Name	Data type	Constraints
1	id	int	Primary key
2	book_title	String	
3	book_author	String	
4	book_summary	String	
5	book_price	int	
6	book_image	List<dynamic>	
7	book_isbn	List<dynamic>	
8	book_total_page	int	
9	book_category	String	
10	total_book	int	

- **Student Table:**

No	Field Name	Data type	Constraints
1	sid	int	Primary key
2	std_rollno	String	
3	std_name	String	
4	std_mobile	int	max_length(10)
5	std_email	String	
6	std_gender	String	
7	std_div	int	
8	std_field	dynamic	
9	std_year	String	
10	std_password	dynamic	

- **Request Table:**

No	Field Name	Data type	Constraints
1	id	int	Primary key
2	book_id	int	
3	student_id	int	
4	book_name	String	

- **Issued Book Table:**

No	Field Name	Data type	Constraints
1	id	int	Primary key
2	book_id	int	
3	student_id	int	
4	book_name	String	
5	issue_date	TimeStamp	
6	return_date	TimeStamp	

- **General Setting Table:**

No		Field Name	Data type	Constraints
1		id	int	Primary key
2		field	Map<String, dynamic>	

User Requirements

6.1 Identification of User Personas:

Student Persona:

- **Name:** Darsh
- **Background:** A university student majoring in English literature, Darsh loves to explore new books and conduct research for her assignments.
- **Goals:** Easily browse and search for books based on genre or author, reserve books for future use, and receive timely notifications about overdue books or available reservations.

Librarian Persona:

- **Name:** Mohit
- **Background:** An experienced librarian with a passion for organising library resources, Mohit is responsible for managing the library's collection and assisting students and faculty.
- **Goals:** Efficiently manage book inventory, track borrowing history, handle user registrations and account management, and generate reports on library usage.

6.2 Use Cases and User Stories:

Use Case 1: Searching for Books

- **User Story:** As a student, Darsh wants to search for books by title, author, or category so that she can find relevant materials for her research paper.

Use Case 2: Placing Book Requests

- **User Story:** As a student, Darsh wants to place a request for a book that is currently unavailable so that she can borrow it once it becomes available.

Use Case 3: Managing User Accounts

- **User Story:** As a librarian, Mohit wants to register new students and faculty members in the library system, assign unique identifiers, and manage account information.

6.3 Functional and Non-functional Requirements:

Functional Requirements:

- **Search Functionality:** Users should be able to search for books by title, author, or category.
- **Reservation System:** Users should be able to reserve books that are currently unavailable.
- **User Account Management:** Librarians should be able to register new users, update user information, and handle account-related issues.
- **Reporting Feature:** Librarians should have access to reports on book circulation, popular genres, and overdue books.

Non-functional Requirements:

- **Performance:** The system should respond quickly to user interactions, ensuring a smooth browsing experience.
- **Security:** User data, including personal information and borrowing history, should be securely stored and accessible only to authorised personnel.

- **Scalability:** The system should be able to accommodate a growing number of users and books without compromising performance.
- **Accessibility :**The user interface should be accessible to users with disabilities, following accessibility guidelines and standards.

Design and Implementation

7.1 User Interface Design Principles:

The user interface design of our Library Management System follows key principles to ensure usability, accessibility, and visual appeal:

- Consistency:

Maintaining consistency in design elements is paramount to providing a cohesive and intuitive user experience. Consistent use of colours, typography, icons, and layout across all screens and interactions helps users navigate the application effortlessly and fosters familiarity. By adhering to a unified design language, we ensure that users can predict how elements will behave and interact throughout the system, enhancing usability and reducing cognitive load.

- Simplicity:

Simplicity is a core tenet of our user interface design philosophy. We embrace a minimalist approach, prioritizing clarity, and ease of use over unnecessary complexity. Streamlined navigation, clear information hierarchy, and minimalist visual elements reduce cognitive load and enable users to focus on their tasks without distraction. By eliminating unnecessary clutter and distractions, we create a clean and uncluttered interface that enhances user engagement and satisfaction.

- Accessibility:

Ensuring accessibility is an integral part of our design process, as we strive to make our Library Management System inclusive and accessible to all users, including those with disabilities. We adhere to accessibility standards such as Web Content Accessibility Guidelines (WCAG) to ensure that our interface is perceivable, operable, and understandable for users with diverse needs. Features such as alternative text for images, keyboard navigation support, and proper contrast ratios enhance accessibility and enable users with disabilities to navigate and interact with the application effectively.

- Feedback:

Providing meaningful feedback is essential for guiding users through their interactions with the system and maintaining engagement. Visual feedback for user actions, such as button presses, form submissions, and loading states, helps users understand the outcome of their actions and reinforces their sense of control and agency. Interactive elements should respond promptly to user input, with subtle animations, transitions, and feedback cues that convey system status and responsiveness. By providing timely and informative feedback, we enhance the user experience and foster a sense of confidence and satisfaction.

- Scalability:

Designing for scalability ensures that our interface remains functional and visually appealing across different screen sizes, resolutions, and devices. We employ responsive design principles and adaptive layouts to accommodate various screen orientations, aspect ratios, and device capabilities. Flexible grid systems, fluid layouts, and scalable UI components enable our interface to adapt gracefully to different viewport sizes without compromising usability or visual coherence. By prioritizing scalability, we future-proof our design and ensure a consistent user experience across a diverse range of devices and platforms.

●

7.2 Screenshots or Wireframes of Key App Screens

●

SYSTEM DESIGN

Library side:

• AUTHENTICATION

10:49



Log in

Student Id

123456

Password

..... 

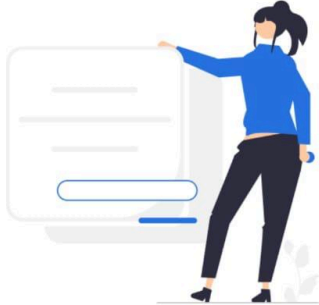
☒ Remember Student Id

☒ Stay Logged In

Register

Don't have an account? [Register](#)

10:54



Please Enter Your Email Address
To Recieve a Email

Email

Enter Email

Send Email

- **BOTTOMBAR**



ALL REQUEST

The Lord of the Rings: The Fellowship of the Ring

From: 20210035

Decline

Accept

ctx

From: 20210035

Decline

Accept

Requests Books Students Profile



ALL BOOK

Search book



Title: The Catcher in the Rye

Author: J.D. Salinger

Category: Fiction

Available book: 30

Total page: 224

Book summary:

Holden Caulfield's journey through New York City after being expelled from school.



Title: To Kill a Mockingbird

Author: Harper Lee

Category: Classic Literature

Available book: 38

Total page: 336

Book summary:

A lawyer defends a black man accused of raping a white woman in the racially divided town of Maycomb.



Requests Books Students Profile

ALL STUDENT

Roll no: 605
Name: vibhu
Email: vibhu@gmail.com
Gender: Female
Mobile: 9316707289
SID: 20210035
Study: FYBCA

Roll no: 667
Name: deep
Email: deep@gmail.com
Gender: Male
Mobile: 9632587412
SID: 21410010
Study: FYBCA

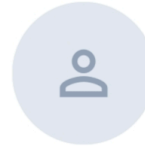
Roll no: 614
Name: Kenisha gajera
Email: keni@gmail.com
Gender: Female
Mobile: 9874563215
SID: 21410030
Study: TYBCA

Roll no: 614
Name: kenisha



Requests Books **Students** Profile

PROFILE

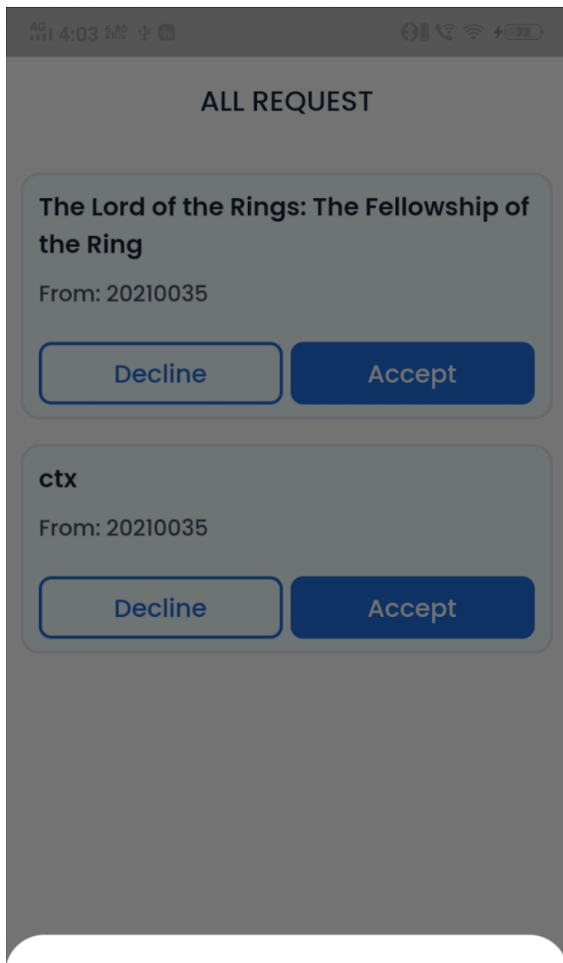


kenigajera@gmail.com

Reset password

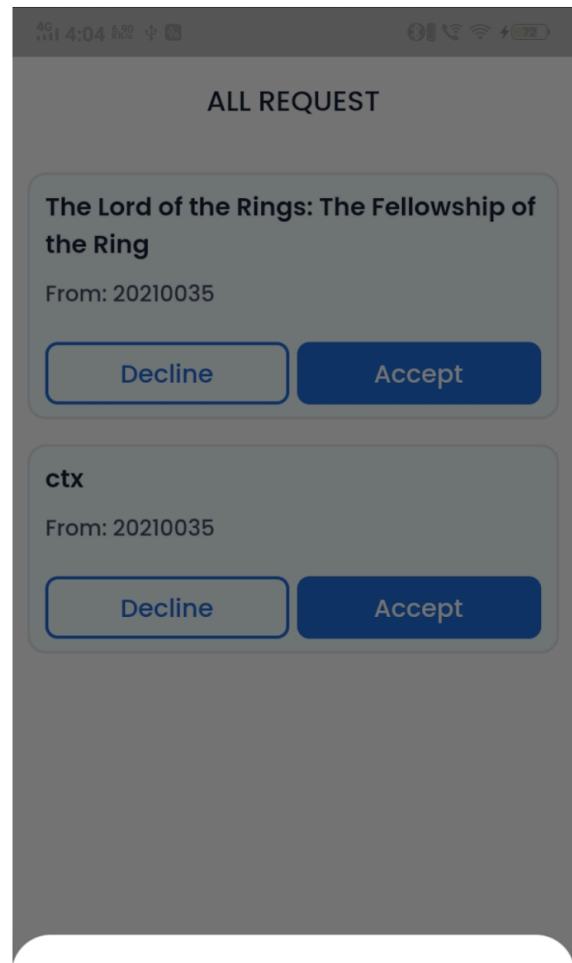
Requests Books Students **Profile**

- **ACCEPT OR DECLINE REQUEST**



Accept the request?

ACCEPT



Decline the request?

DECLINE

• ADD OR DELETE BOOK

4G
4:28
2.30
79

<
ADD BOOK

Book Title*
Enter book title

Book Summary*
Please enter your summary here

Author of the Book*
Enter author name

Category*
Enter category

Total page*
Enter total page

Price*
Total Books*
Enter price
0

Book ISBN*
Add Name
+

Book Images*
+

ADD BOOK

4G
4:06
8.40
72

<
EDIT BOOK

Book Title*
The Catcher in the Rye

Book Summary*
Holden Caulfield's journey through New York City after being expelled from school.

Author of the Book*
J.D. Salinger

Category*
Fiction

Total page*
224

Price*
Total Books*
12.99
30

Book ISBN*
978-0316769174
+

Book Images*




EDIT BOOK

• SEARCH BOOK WITH BOOK TITLE



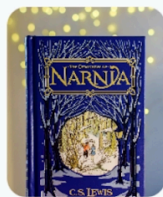
ALL BOOK



Title: The Catcher in the Rye
Author: J.D. Salinger
Category: Fiction
Available book: 30
Total page: 224

Book summary:

Holden Caulfield's journey through New York City after being expelled from school.



Title: To Kill a Mockingbird
Author: Harper Lee
Category: Classic Literature
Available book: 38
Total page: 336

Book summary:

A lawyer defends a black man accused of raping a white woman in the racially divided town of Maycomb.



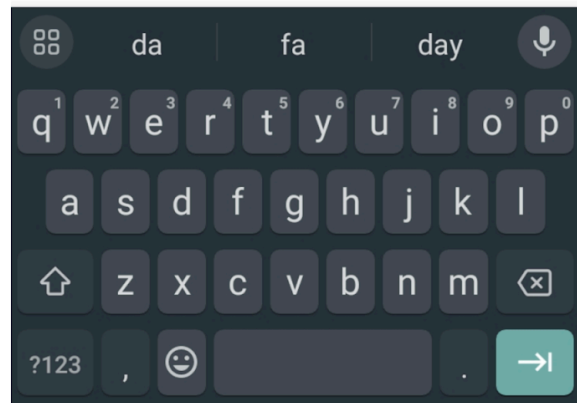
- Requests
- Books
- Students
- Profile



ALL BOOK



Title: The Da Vinci Code
Author: Dan Brown
Category: Mystery
Available book: 22
Total page: 592
Book summary:
 A professor and a cryptologist race to uncover a secret that could shake the foundations of Christianity.



- **ADD OR DELETE BOOK**

4G 4:42 100% 5G 100% 100%

< ADD STUDENT

Name
Enter Your Name

Select Your gender
Male

Email Address
Email Address

Mobile Number
Enter Mobile Number

Student Id
Enter Student Id

Class
BCA FYBCA 1

Roll Number
Enter Roll Number

Password
Enter Password

ADD

4G 4:06 100% 5G 100% 100%

< EDIT STUDENT

Name
deep

Select Your gender
Male

Email Address
deep@gmail.com

Mobile Number
9632587412

Student Id
21410010

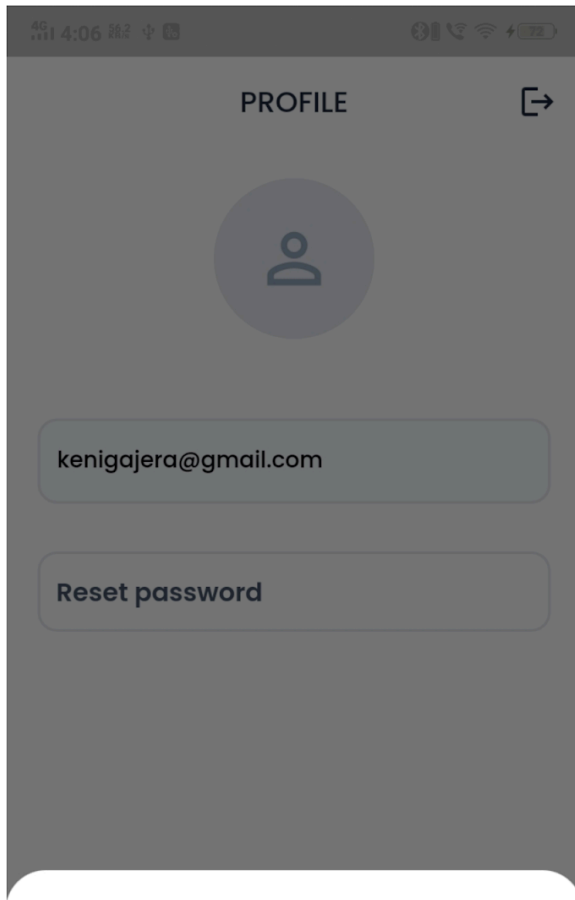
Class
BCA FYBCA 1

Roll Number
667

Password
1234567

EDIT

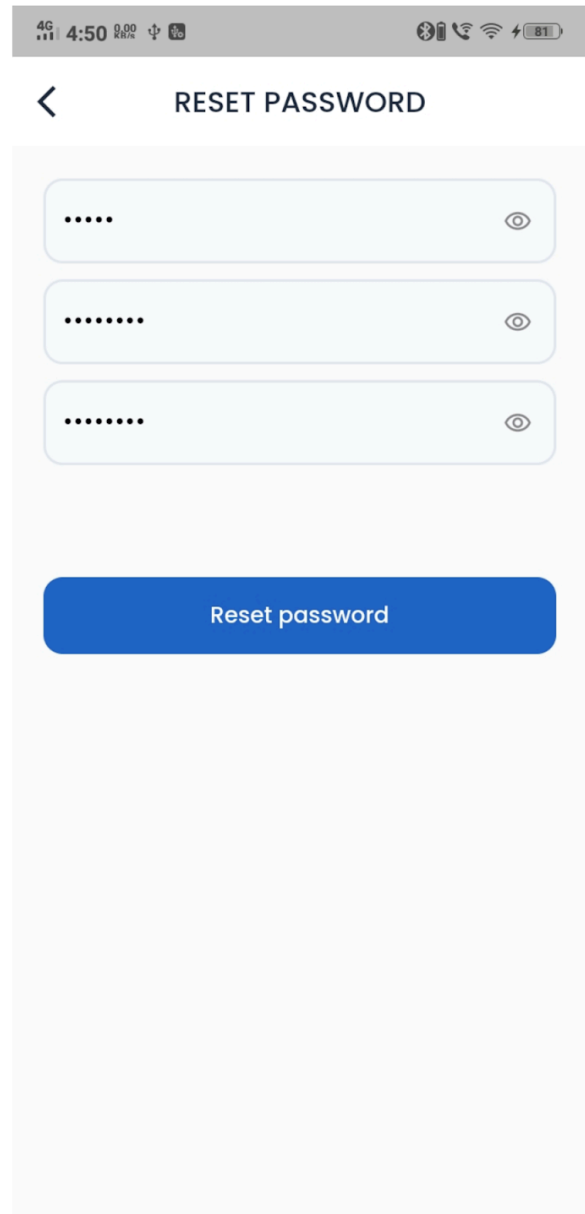
- **LOGOUT AND RESET PASSWORD**



Logout

are you sure you want to logout

Logout



- **Student side:**

- **AUTHENTICATION**

10:49 10:49



Log in **Log in**

Student Id Student Id

123456 Enter Student Id

Password Password

..... Enter Password

☒ Remember Student Id ☐ Remember Student Id

☒ Stay Logged In ☐ Stay Logged In

Register Register

Don't have an account? [Register](#) Don't have an account? [Register](#)

10:50

<



Create Your Account

Enter Your Name

Enter Your Name

Select Your gender

Male

Email Address

Enter Email Address

Mobile Number

Enter Mobile Number

Student Id

Enter Student Id

Class

Roll Number

Enter Roll Number

Password

Enter Password

Confirm Password

Enter Confirm Password

10:52

<



Create Your Account

Enter Your Name

Urmit Goyani

Select Your gender

Male

Email Address

urmit.goyani.mscit2021@aurouniversity.edu.in

Mobile Number

9016223692

Student Id

123456

● BOTTOMBAR

16:16

ALL BOOKS

 Search book



Title: The Catcher in the Rye
Author: J.D. Salinger
Category: Fiction
Available book: 30
Total page: 224

Book summary:
Holden Caulfield's journey through New York City after being expelled from school.

REQUEST BOOK



Title: To Kill a Mockingbird
Author: Harper Lee
Category: Classic Literature
Available book: 38
Total page: 336

Book summary:
A lawyer defends a black man accused of raping a white woman in the racially divided town of Maycomb.

 Dashboard

 Requests

 Account

16:16

BOOK REQUESTS

Your request for The Lord of the Rings: The Fellowship of the Ring has been accepted

Issued Date: 22/09/2023
Return Date: 07/10/2023

Your request for To Kill a Mockingbird has been accepted

Issued Date: 22/09/2023
Return Date: 07/10/2023

Your request for The Catcher in the Rye has been accepted

Issued Date: 22/09/2023
Return Date: 07/10/2023

Your request for To Kill a Mockingbird has been accepted

Issued Date: 22/09/2023
Return Date: 07/10/2023

Your request for yyyy has been accepted

Issued Date: 22/09/2023

 Dashboard

 Requests

 Account

16:16



vibhu

vibhu@gmail.com

Edit Profile

Mobile Number

9316707289

Gender

Female

Field

BCA

Class

FYBCA

Division

1

Roll Number

605

Student Id

20210035

Log Out

Dashboard

Requests

Account

16:16

ALL BOOKS

Search book



Title: The Catcher in the Rye

Author: J.D. Salinger

Category: Fiction

Available book: 30

Total page: 224

Are you sure to send a book request ?

Cancel

Yes



Title: To Kill a Mockingbird

Author: Harper Lee

Category: Classic Literature

Available book: 38

Total page: 336

Book summary:

A lawyer defends a black man accused of raping a white woman in the racially divided town of Maycomb.

Dashboard

Requests

Account

7.3 Implementation Approach for Major Features

1. User Authentication:

- **Implementation:**

We utilize Firebase Authentication to handle user authentication securely, supporting various authentication methods such as email/password, phone number, and social authentication (e.g., Google Sign-In).

- **Approach:**

We implement user authentication using Firebase Authentication SDK, integrating authentication flows seamlessly into the application's login and registration screens. Firebase Authentication provides robust security features, including secure password storage, identity verification, and multi-factor authentication, ensuring that user accounts are protected against unauthorized access and malicious attacks.

2. Real-time Database:

- **Implementation:**

We leverage Firebase Realtime Database to store and synchronize real-time data related to user accounts, book records, borrowing history, and request statuses.

- **Approach:**

Using Firebase Realtime Database SDK, we integrate real-time data synchronization capabilities into the application, allowing users to view and update data in real-time across multiple devices. We structure data in a denormalized format optimized for real-time updates and efficient querying, minimizing data duplication and maximizing scalability. Firebase Database Rules are configured to enforce data validation and access control, ensuring that only authorized users can read and write data to the database.

3. Cloud Storage:

- **Implementation:**

We utilize Firebase Cloud Storage to store and manage multimedia assets such as book covers, author photos, and user profile pictures.

- **Approach:**

Using Firebase Storage SDK, we integrate cloud storage functionality into the application, enabling users to upload, download, and manage files securely in the cloud. We implement client-side and server-side security rules to restrict access to storage buckets and ensure data integrity and confidentiality. Firebase Storage enables seamless integration with other Firebase services, such as Firebase Authentication and Firebase Realtime Database, to provide a unified storage solution for the application.

4. Firestore:

- **Implementation:**

We leverage Cloud Firestore, Firebase's NoSQL document database, to store structured data and perform complex queries.

- **Approach:**

Using Firestore SDK, we model data using collections and documents, organizing data hierarchically to facilitate efficient querying and data retrieval. We utilize Firestore's powerful query capabilities to filter, sort, and paginate data in real-time, enabling dynamic and responsive user experiences. Firestore's scalability and performance optimizations ensure that the application can handle large volumes of data and high concurrency without sacrificing performance or reliability.

5. Device Alerts:

- **Implementation:**

-

We implement device alerts to provide users with timely notifications and reminders about important events, such as due dates, overdue fines, and book availability.

- **Approach:**

Device alerts are triggered within the application itself based on predefined conditions or events, such as upcoming due dates or changes in request statuses. We utilize local notification APIs provided by the platform (e.g., Android's AlarmManager or iOS's UNNotificationService) to schedule and display alerts directly on the user's device. By leveraging platform-specific APIs, we ensure that device alerts are personalized, actionable, and respectful of user preferences, allowing users to stay informed and engaged with the application while minimizing disruptions.

Testing and Quality Assurance

8.1 Overview of Testing Methodologies:

- **Unit Testing:**

Unit testing focuses on testing individual components or functions in isolation to verify that they behave as expected. This methodology ensures that each unit of code performs its intended functionality correctly, independent of other parts of the system. We employ tools like Flutter's built-in test framework or third-party libraries like Mockito for mocking dependencies and facilitating unit testing. By writing comprehensive unit tests, we can identify and fix bugs early in the development process, improving code quality and maintainability.

- **Integration Testing:**

Integration testing verifies interactions between different modules or subsystems to ensure they work together correctly as a cohesive system. This methodology tests database interactions, API integrations, and UI components integration to validate end-to-end functionality across system boundaries. We design integration tests to cover critical paths and edge cases, ensuring comprehensive test coverage and detecting integration issues early in the development lifecycle. By validating system integrations, we mitigate risks associated with component dependencies and ensure seamless interoperability between system components.

- **End-to-End Testing:**

End-to-end testing validates the entire system's functionality from the user's perspective, simulating real-world user interactions and scenarios. This methodology tests user flows, navigation paths, and user interface behaviors to ensure that the application functions correctly and meets user expectations. We automate end-to-end tests using tools like Flutter Driver or Firebase Test Lab for mobile apps, executing scripted test scenarios across multiple devices and platforms. By automating repetitive test scenarios, we improve test efficiency and coverage, enabling faster feedback cycles and more reliable releases.

- **Usability Testing:**

Usability testing evaluates the app's ease of use, intuitiveness, and overall user experience from the perspective of real users. This methodology gathers feedback from target users through user interviews, surveys, and observation sessions, identifying usability issues and areas for improvement. We involve representative users throughout the development process, soliciting feedback at various stages to validate design decisions and iterate on user interface designs. By prioritizing usability testing, we ensure that the Library Management System is user-friendly, accessible, and aligned with user needs and expectations.

- **Performance Testing:**

Performance testing assesses the system's responsiveness, scalability, and stability under various load conditions. This methodology measures key performance metrics such as response times, throughput, and resource utilization to identify performance bottlenecks and optimize system performance. We use tools like JMeter or Firebase Performance Monitoring to simulate user traffic and monitor system performance in real-time. By conducting performance testing, we ensure that the Library Management System can handle anticipated user loads and maintain optimal performance under peak usage conditions.

8.2 Test Cases and Scenarios:

Search Functionality Test Case:

- **Scenario:** User searches for a book by title.
- **Expected Outcome:** The app displays relevant search results matching the book title entered by the user.

User Registration Test Case:

- **Scenario:** User registers for a new account.

- **Expected Outcome:** User receives a verification email and successfully completes the registration process.

Book Reservation Test Case:

- **Scenario:** User reserves a book that is currently available.
- **Expected Outcome:** The book is marked as reserved, and the user receives a confirmation notification.

8.3 Results of Testing Phase:

- **Unit Testing:**

Unit testing focuses on testing individual components or functions in isolation to verify that they behave as expected. This methodology ensures that each unit of code performs its intended functionality correctly, independent of other parts of the system. We employ tools like Flutter's built-in test framework or third-party libraries like Mockito for mocking dependencies and facilitating unit testing. By writing comprehensive unit tests, we can identify and fix bugs early in the development process, improving code quality and maintainability.

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Deployment

9.1 Deployment Environment:

- **Hosting Platform:**

The library app will be deployed on a cloud hosting platform such as Firebase Hosting or AWS Amplify to ensure scalability, reliability, and ease of management. Cloud hosting offers benefits such as automatic scaling, global distribution, and built-in security features.

- **Backend Services:**

The app's backend services, including Firebase Firestore for data storage and Firebase Authentication for user authentication, will be hosted on Firebase's cloud infrastructure. Firebase provides a fully managed backend platform with real-time database capabilities, user authentication, and serverless functions for handling business logic.

- **Frontend Framework:**

The Flutter frontend will be compiled into platform-specific binaries (APK for Android, IPA for iOS) and deployed to respective app stores for mobile platforms. For web deployment, the Flutter web build will be hosted on the chosen cloud platform. Flutter's cross-platform capabilities enable code sharing between mobile and web platforms, streamlining the deployment process.

9.2 Steps Involved in Deploying the App:

- **Prepare Production Builds:**

Compile the Flutter frontend into production-ready builds for Android, iOS, and web platforms using Flutter CLI commands. Optimize build configurations for performance and size considerations.

- **Set up Firebase Project:**

Configure Firebase project settings, including enabling Firebase Hosting and linking the project to Firebase services such as Firestore and Authentication. Set up Firebase project environments for development, staging, and production.

- **Configure Firebase Hosting:**

Set up Firebase Hosting by installing Firebase CLI, initializing the project, and deploying the Flutter web build to Firebase Hosting using CLI commands. Configure hosting settings such as custom domains, redirects, and cache control policies.

- **Configure Domain and SSL:**

If a custom domain and SSL are required, configure domain settings and enable SSL certificates for secure HTTPS connections. Configure DNS settings to point the custom domain to Firebase Hosting.

- **Deploy Mobile Apps:**

Upload the compiled APK and IPA files to Google Play Store and Apple App Store, respectively, following their submission guidelines and review processes. Prepare app metadata, screenshots, and promotional assets for app store listings.

- **Test Deployment:**

Conduct thorough testing of the deployed app across different devices and platforms to ensure compatibility and functionality. Perform regression testing to validate app behavior in production environments.

- **Monitor Performance:**

Monitor app performance, user feedback, and analytics data using Firebase Analytics and other monitoring tools to identify any

issues and optimize performance. Set up alerts and notifications for critical performance metrics such as error rates and response times.

9.3 Challenges Encountered During Deployment:

- **Cross-platform Compatibility:**

Ensuring consistent behavior and performance across different platforms (Android, iOS, web) can be challenging due to platform-specific differences and limitations. Compatibility testing on various devices and browsers is essential to identify and address platform-specific issues.

- **Configuration Errors:**

Configuring Firebase services, domain settings, SSL certificates, and deployment pipelines requires careful attention to detail to avoid configuration errors and deployment failures. Documentation and automation tools can help streamline the deployment process and minimize human errors.

- **App Store Submission:**

Submitting the app to Google Play Store and Apple App Store involves adhering to strict guidelines and passing app review processes, which may require addressing compliance issues and meeting platform-specific requirements. App store submission timelines and review processes can vary, impacting deployment schedules.

- **Performance Optimization:**

Optimizing app performance and minimizing loading times on different devices and network conditions is an ongoing challenge that requires continuous monitoring and optimization efforts. Performance optimization techniques such as code splitting, lazy loading, and image optimization can improve app responsiveness and user experience.

Maintenance and Future Enhancements

10.1 Strategies for Ongoing Maintenance:

- **Regular Updates:**

Schedule regular updates and maintenance checks to ensure the app remains up-to-date with the latest security patches, bug fixes, and feature enhancements. Implement an agile development process with iterative releases to address user feedback and evolving requirements.

- **Monitoring and Analytics:**

Implement monitoring tools and analytics to track app performance, user engagement, and feedback. Utilize tools such as Firebase Analytics or Google Analytics to gather insights into user behavior, app usage patterns, and performance metrics. Use this data to identify areas for improvement and prioritize maintenance tasks.

- **User Support:**

Provide ongoing user support through various channels such as email, in-app chat support, or a dedicated support portal to address user queries, issues, and feature requests promptly. Establish service level agreements (SLAs) for response times and resolution times to ensure timely support delivery.

- **Documentation and Training:**

Maintain comprehensive documentation for developers, administrators, and end-users to facilitate troubleshooting, configuration, and usage of the app. Offer training sessions or tutorials for new features or updates to empower users to maximize the app's capabilities.

10.2 Feedback Mechanisms for Users:

- **In-App Feedback:**

Incorporate in-app feedback mechanisms such as surveys, rating prompts, or feedback forms to gather user input and suggestions directly within the app. Prompt users for feedback at key touchpoints to capture their insights and opinions.

- **User Forums or Communities:**

Create online forums or communities where users can discuss their experiences, share feedback, and vote on feature requests. Engage with the community to address concerns, answer questions, and prioritize enhancements based on user input.

- **User Interviews and Focus Groups:**

Conduct user interviews and focus groups to gather qualitative feedback and insights into user needs, pain points, and preferences. Use techniques such as persona development and journey mapping to understand user personas and tailor the app experience accordingly.

10.3 Potential Future Features and Improvements:

- **Advanced Search Functionality:**

Enhance the search capabilities with advanced filters, predictive search suggestions, and personalized recommendations based on user preferences and browsing history. Implement natural language processing (NLP) techniques to improve search accuracy and relevance.

- **Offline Access:**

Implement offline access capabilities, allowing users to access cached content and perform basic tasks even when offline. Utilize technologies such as service workers and local storage to enable seamless offline experiences and sync data with the server when connectivity is restored.

- **Gamification Elements:**

Introduce gamification elements such as badges, achievements, or leaderboards to incentivize user engagement, encourage exploration of library resources, and promote learning. Reward users for completing tasks, achieving milestones, or participating in community activities.

- **Accessibility Enhancements:**

Continuously improve accessibility features to ensure the app is inclusive and usable by users with disabilities. Adhere to accessibility standards such as WCAG (Web Content Accessibility Guidelines) and conduct regular accessibility audits to identify and address usability barriers.

- **Integration with Learning Management Systems:**

Integrate with existing learning management systems used by educational institutions to streamline access to library resources and support seamless integration with course materials. Enable single sign-on (SSO) and interoperability with LMS platforms such as Moodle, Canvas, or Blackboard.

Conclusion

11.1 Summary of Key Findings:

- Through meticulous requirement gathering and analysis, we identified the needs of both students and library staff, leading to the development of a comprehensive Library Management System.
- The system design phase resulted in intuitive user interfaces and efficient backend architecture, leveraging Flutter for the frontend and Firebase for the backend.
- Deployment on cloud platforms like Firebase Hosting ensured scalability, reliability, and accessibility across various devices and platforms.
- Testing and quality assurance efforts validated the functionality, usability, and performance of the app, ensuring a seamless user experience.

11.2 Achievements and Challenges Faced:

Achievements:

- Successful development of a modern and efficient Library Management System catering to the needs of students and library staff.
- Seamless integration of Flutter frontend with Firebase backend, ensuring real-time data synchronisation and secure user authentication.
- Positive feedback from users during testing phases, validating the usability and effectiveness of the app.

Challenges Faced:

- Addressing platform-specific differences and limitations during cross-platform development for Android, iOS, and web.

- Configuring and deploying the app to cloud hosting platforms, overcoming configuration errors and compatibility issues.
- Balancing feature requirements with project constraints, prioritising key functionalities while managing scope creep.

11.3 Lessons Learned:

- Effective requirement gathering and analysis are crucial for defining project scope and objectives accurately, ensuring alignment with user needs and expectations.
- Continuous communication and collaboration among team members are essential for successful project execution, facilitating problem-solving and decision-making processes.
- Regular testing and quality assurance are integral parts of the development lifecycle, helping identify and address issues early, minimising risks and ensuring high-quality deliverables.
- Embracing feedback and lessons learned from each phase of the project enables continuous improvement and innovation, driving future success and growth.

In conclusion, the development of the Library Management System has been a rewarding journey, marked by achievements, challenges, and valuable lessons learned. By leveraging best practices, collaboration, and dedication, we have delivered a solution that empowers users, enhances efficiency, and contributes to the advancement of academic endeavours.

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8. Material Design. (n.d.). Retrieved from <https://material.io/design>
9. Cupertino Widgets. (n.d.). Retrieved from <https://flutter.dev/docs/development/ui/widgets/cupertino>